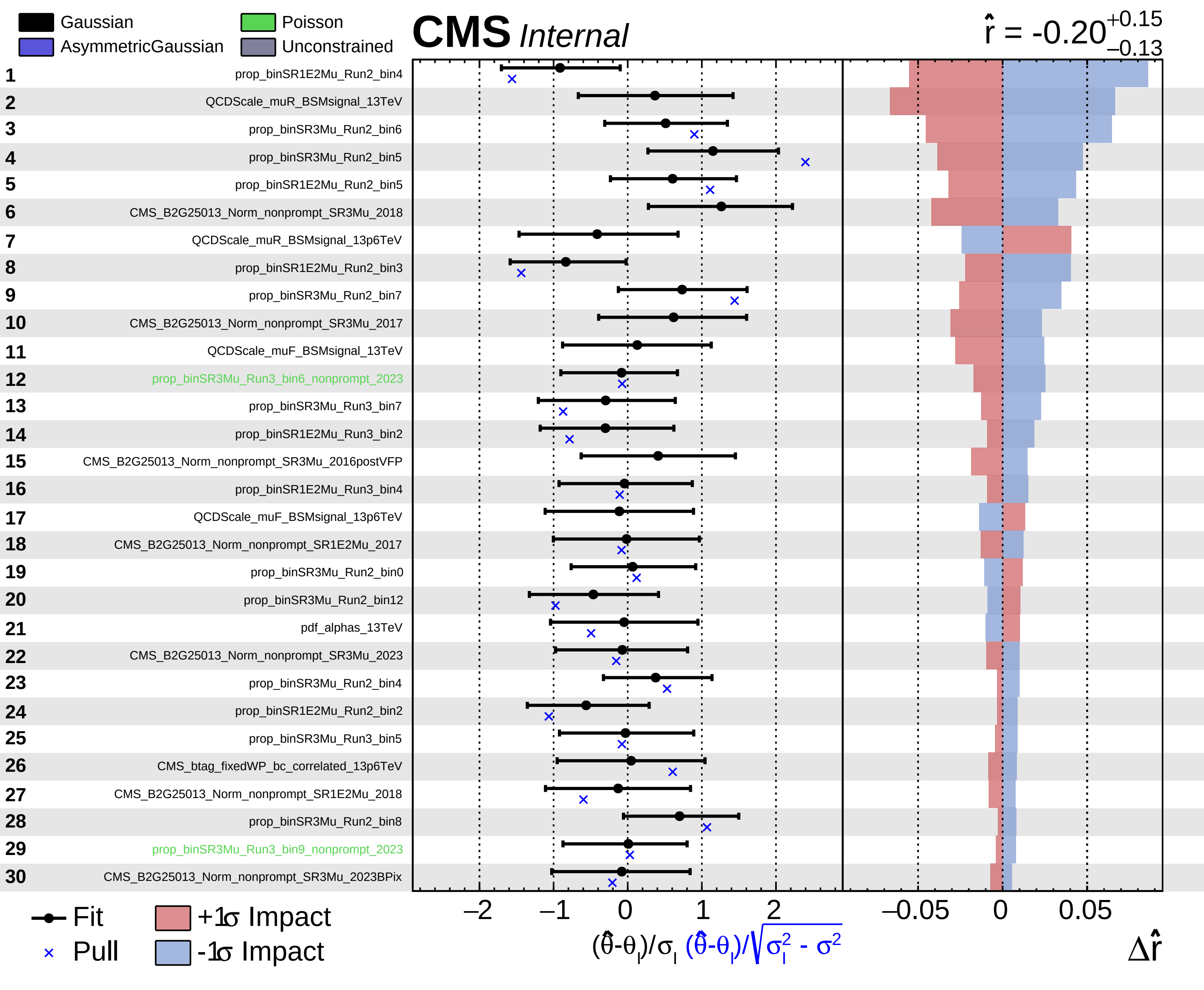
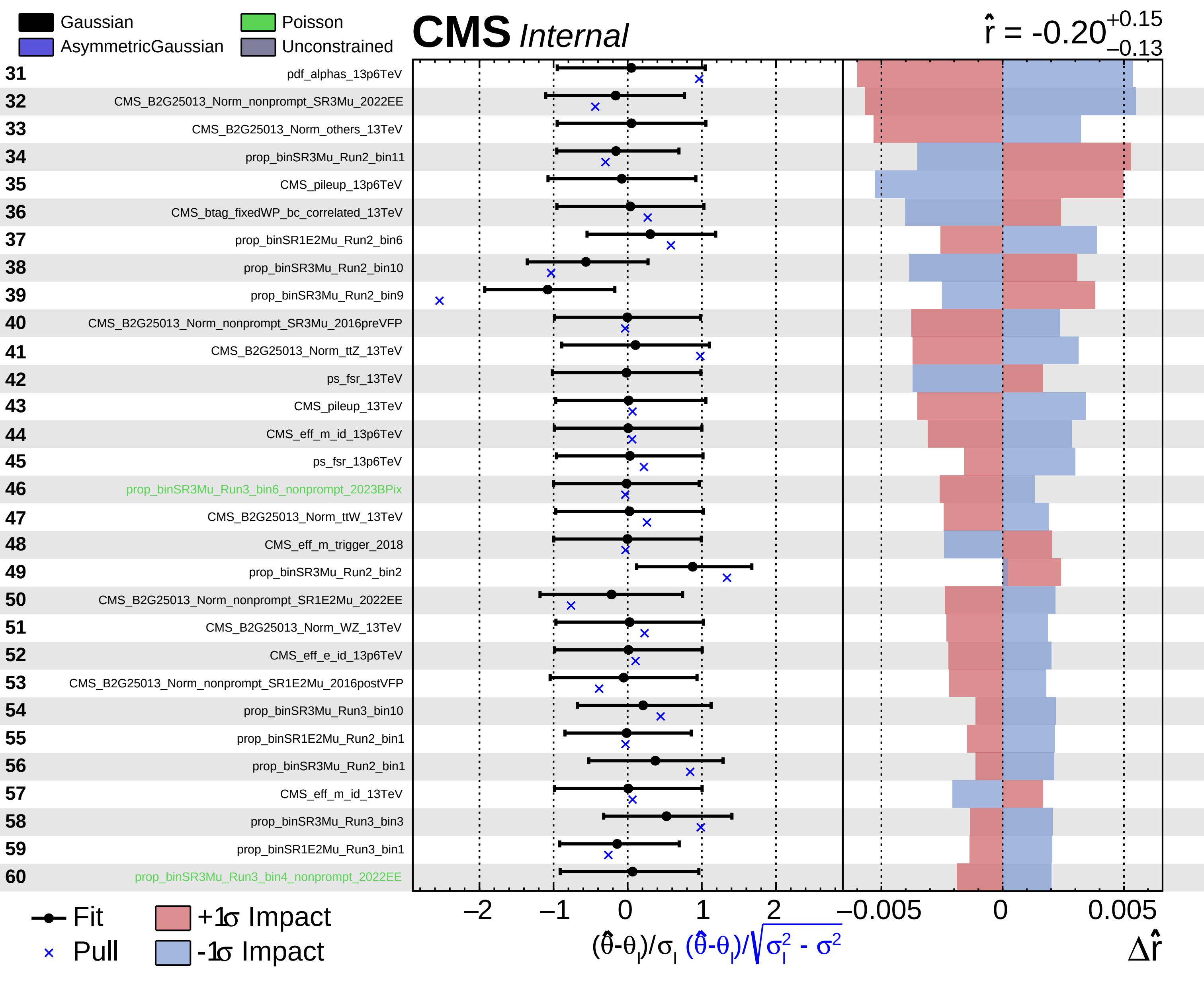
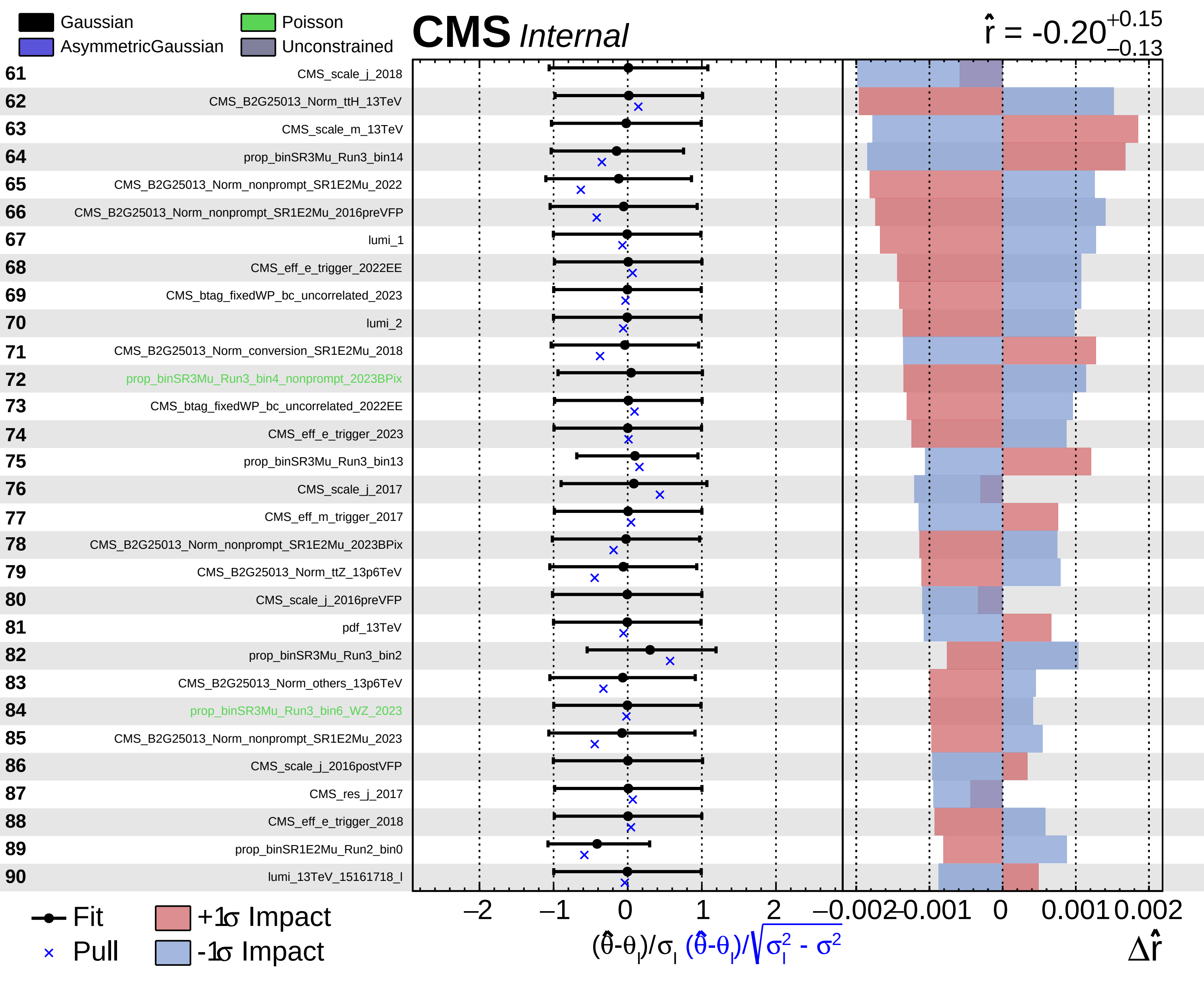
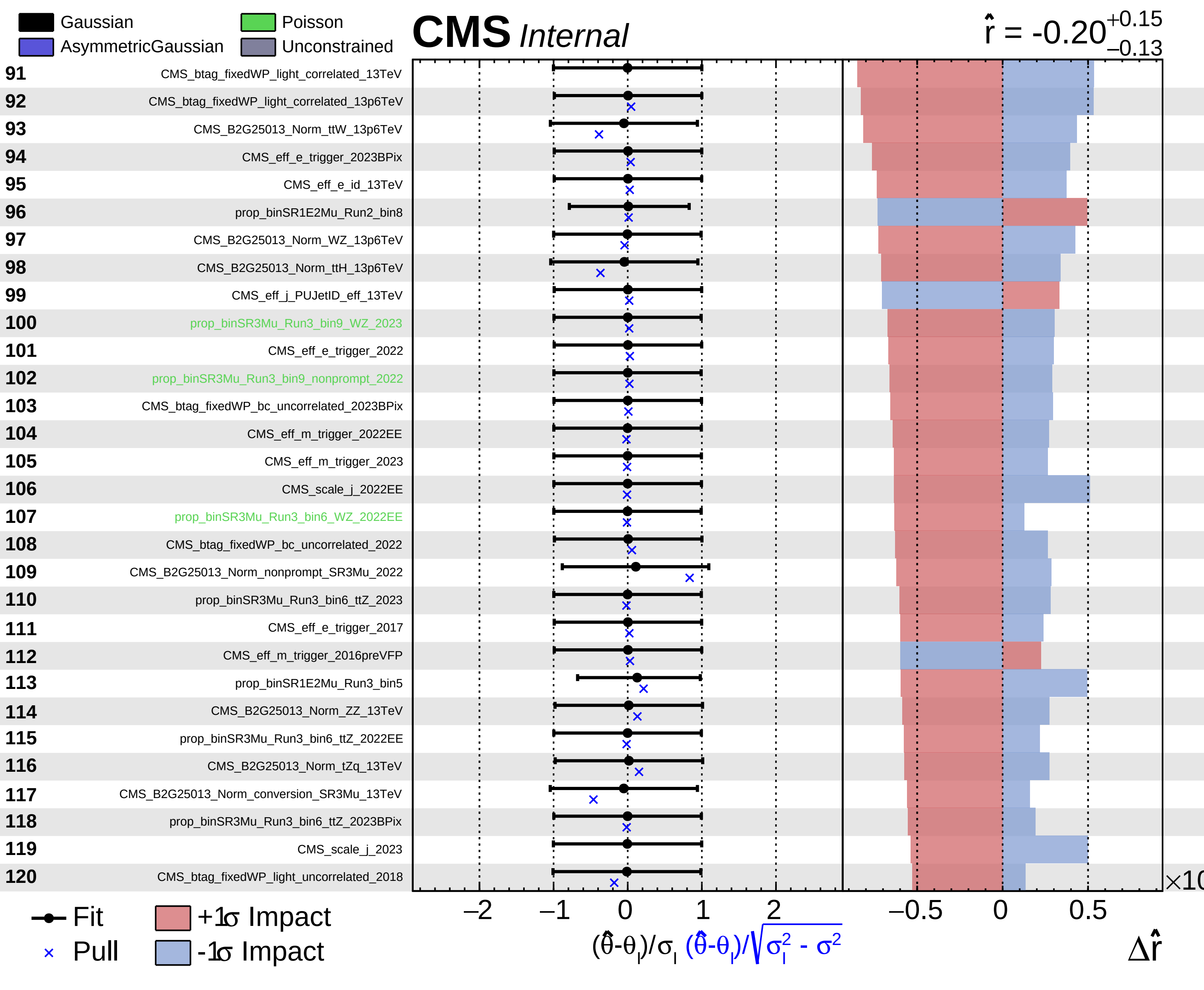


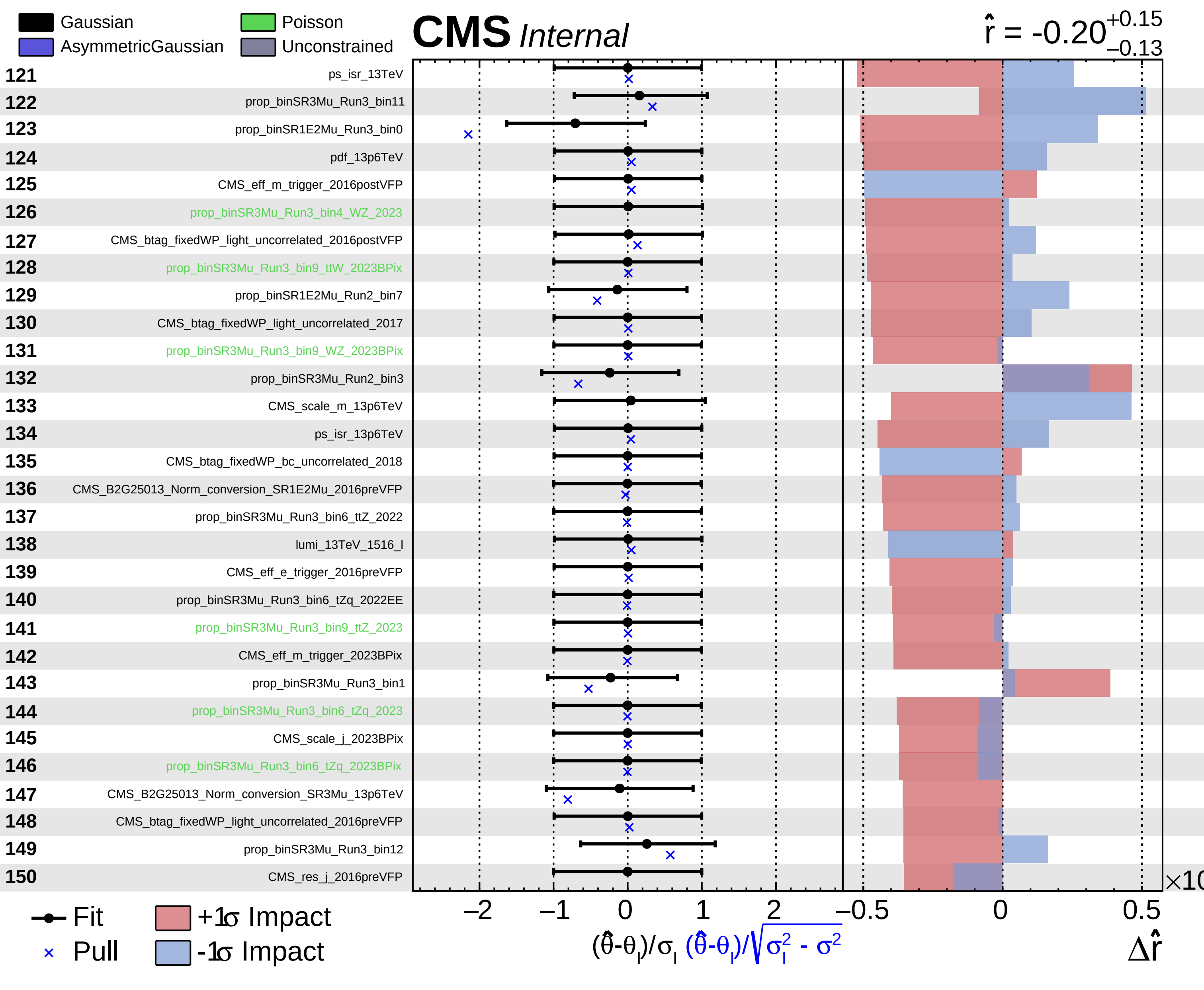
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = -0.20^{+0.15}_{-0.13}$
1 prop_binSR1E2Mu_Run2_bin4
2 QCDScale_muR_BSMsignal_13TeV
3 prop_binSR3Mu_Run2_bin6
4 prop_binSR3Mu_Run2_bin5
5 prop_binSR1E2Mu_Run2_bin5
6 CMS_B2G25013_Norm_nonprompt_SR3Mu_2018
7 QCDScale_muR_BSMsignal_13p6TeV
8 prop_binSR1E2Mu_Run2_bin3
9 prop_binSR3Mu_Run2_bin7
10 CMS_B2G25013_Norm_nonprompt_SR3Mu_2017
11 QCDScale_muF_BSMsignal_13TeV
12 prop_binSR3Mu_Run3_bin6_nonprompt_2023
13 prop_binSR3Mu_Run3_bin7
14 prop_binSR1E2Mu_Run3_bin2
15 CMS_B2G25013_Norm_nonprompt_SR3Mu_2016postVFP
16 prop_binSR1E2Mu_Run3_bin4
17 QCDScale_muF_BSMsignal_13p6TeV
18 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2017
19 prop_binSR3Mu_Run2_bin0
20 prop_binSR3Mu_Run2_bin12
21 pdf_alphas_13TeV
22 CMS_B2G25013_Norm_nonprompt_SR3Mu_2023
23 prop_binSR3Mu_Run2_bin4
24 prop_binSR1E2Mu_Run2_bin2
25 prop_binSR3Mu_Run3_bin5
26 CMS_btag_fixedWP_bc_correlated_13p6TeV
27 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2018
28 prop_binSR3Mu_Run2_bin8
29 prop_binSR3Mu_Run3_bin9_nonprompt_2023
30 CMS_B2G25013_Norm_nonprompt_SR3Mu_2023BPix
−2
−1
0
1
2
−0.05
0
0.05
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = -0.20^{+0.15}_{-0.13}$
31 pdf_alphas_13p6TeV
32 CMS_B2G25013_Norm_nonprompt_SR3Mu_2022EE
33 CMS_B2G25013_Norm_others_13TeV
34 prop_binSR3Mu_Run2_bin11
35 CMS_pileup_13p6TeV
36 CMS_btag_fixedWP_bc_correlated_13TeV
37 prop_binSR1E2Mu_Run2_bin6
38 prop_binSR3Mu_Run2_bin10
39 prop_binSR3Mu_Run2_bin9
40 CMS_B2G25013_Norm_nonprompt_SR3Mu_2016preVFP
41 CMS_B2G25013_Norm_ttZ_13TeV
42 ps_fsr_13TeV
43 CMS_pileup_13TeV
44 CMS_eff_m_id_13p6TeV
45 ps_fsr_13p6TeV
46 prop_binSR3Mu_Run3_bin6_nonprompt_2023BPix
47 CMS_B2G25013_Norm_ttW_13TeV
48 CMS_eff_m_trigger_2018
49 prop_binSR3Mu_Run2_bin2
50 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2022EE
51 CMS_B2G25013_Norm_WZ_13TeV
52 CMS_eff_e_id_13p6TeV
53 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2016postVFP
54 prop_binSR3Mu_Run3_bin10
55 prop_binSR1E2Mu_Run2_bin1
56 prop_binSR3Mu_Run2_bin1
57 CMS_eff_m_id_13TeV
58 prop_binSR3Mu_Run3_bin3
59 prop_binSR1E2Mu_Run3_bin1
60 prop_binSR3Mu_Run3_bin4_nonprompt_2022EE
−2 −1 0 1 2
−0.005 0 0.005
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

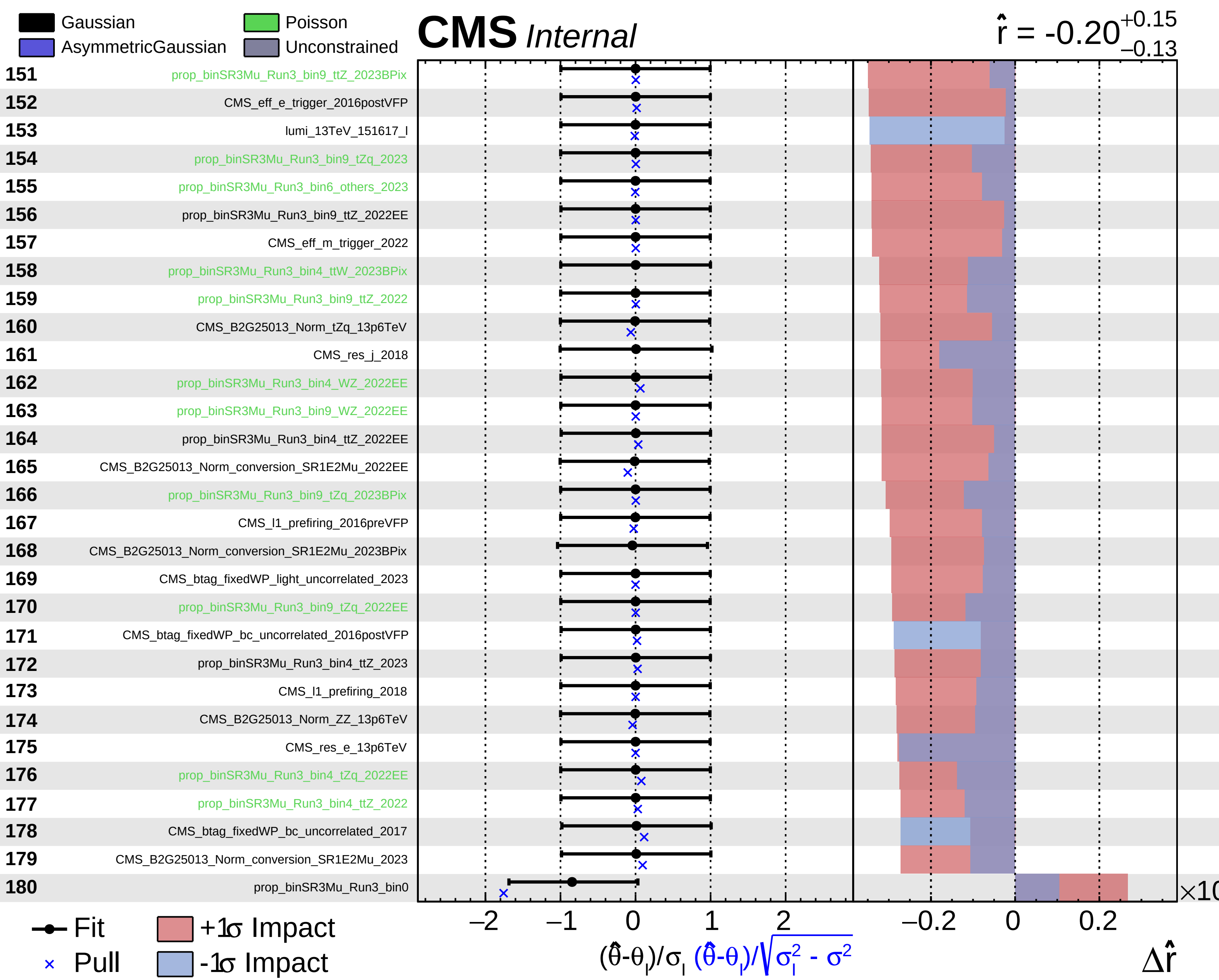


Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = -0.20 +0.15 −0.13
61 CMS_scale_j_2018
62 CMS_B2G25013_Norm_ttH_13TeV
63 CMS_scale_m_13TeV
64 prop_binSR3Mu_Run3_bin14
65 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2022
66 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2016preVFP
67 lumi_1
68 CMS_eff_e_trigger_2022EE
69 CMS_btag_fixedWP_bc_uncorrelated_2023
70 lumi_2
71 CMS_B2G25013_Norm_conversion_SR1E2Mu_2018
72 prop_binSR3Mu_Run3_bin4_nonprompt_2023BPix
73 CMS_btag_fixedWP_bc_uncorrelated_2022EE
74 CMS_eff_e_trigger_2023
75 prop_binSR3Mu_Run3_bin13
76 CMS_scale_j_2017
77 CMS_eff_m_trigger_2017
78 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2023BPix
79 CMS_B2G25013_Norm_ttZ_13p6TeV
80 CMS_scale_j_2016preVFP
81 pdf_13TeV
82 prop_binSR3Mu_Run3_bin2
83 CMS_B2G25013_Norm_others_13p6TeV
84 prop_binSR3Mu_Run3_bin6_WZ_2023
85 CMS_B2G25013_Norm_nonprompt_SR1E2Mu_2023
86 CMS_scale_j_2016postVFP
87 CMS_res_j_2017
88 CMS_eff_e_trigger_2018
89 prop_binSR1E2Mu_Run2_bin0
90 lumi_13TeV_15161718_I
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$

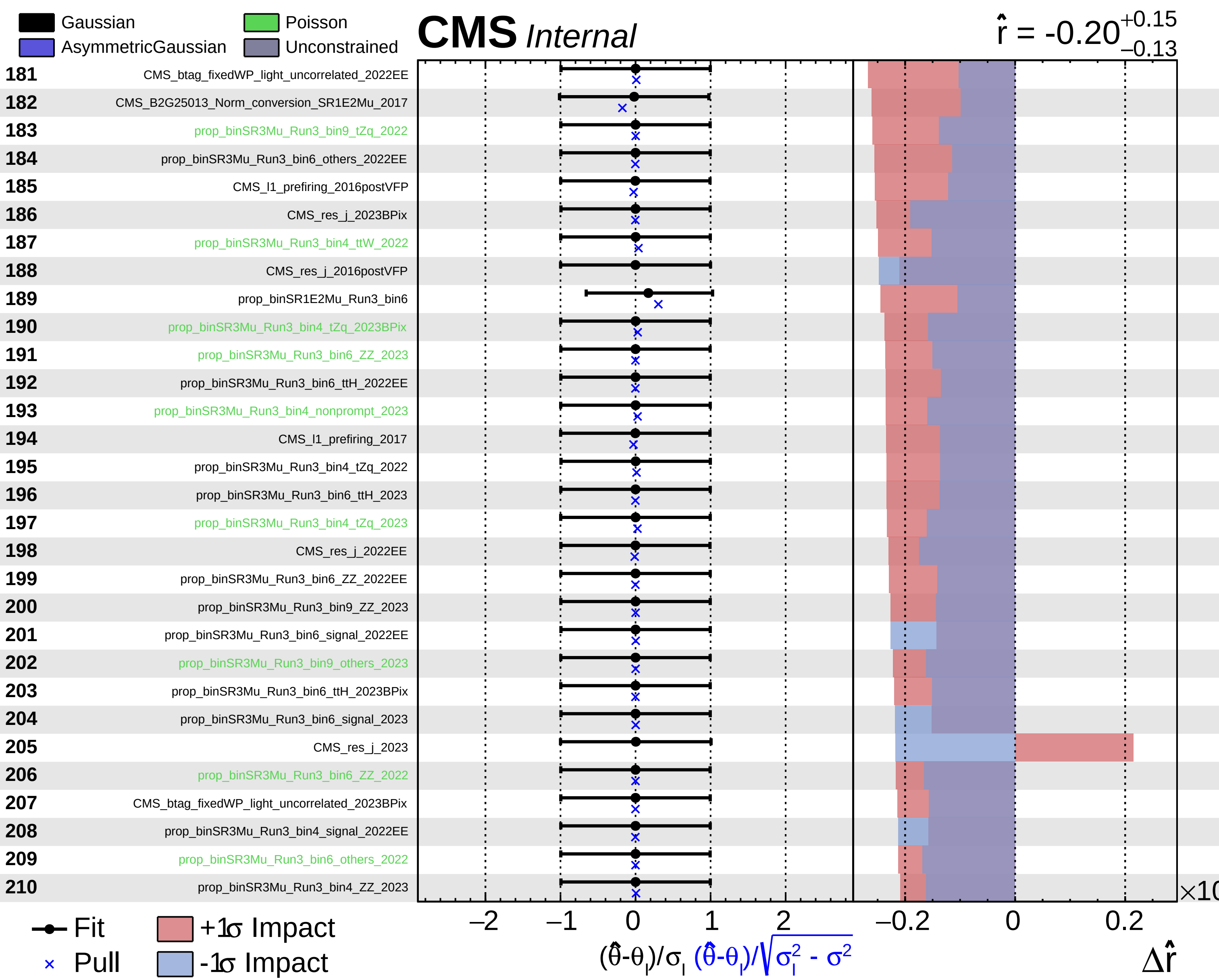

Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Internal
$\hat{r} = -0.20^{+0.15}_{-0.13}$
91 CMS_btag_fixedWP_light_correlated_13TeV
92 CMS_btag_fixedWP_light_correlated_13p6TeV
93 CMS_B2G25013_Norm_ttW_13p6TeV
94 CMS_eff_e_trigger_2023BPix
95 CMS_eff_e_id_13TeV
96 prop_binSR1E2Mu_Run2_bin8
97 CMS_B2G25013_Norm_WZ_13p6TeV
98 CMS_B2G25013_Norm_ttH_13p6TeV
99 CMS_eff_j_PUJetID_eff_13TeV
100 prop_binSR3Mu_Run3_bin9_WZ_2023
101 CMS_eff_e_trigger_2022
102 prop_binSR3Mu_Run3_bin9_nonprompt_2022
103 CMS_btag_fixedWP_bc_uncorrelated_2023BPix
104 CMS_eff_m_trigger_2022EE
105 CMS_eff_m_trigger_2023
106 CMS_scale_j_2022EE
107 prop_binSR3Mu_Run3_bin6_WZ_2022EE
108 CMS_btag_fixedWP_bc_uncorrelated_2022
109 CMS_B2G25013_Norm_nonprompt_SR3Mu_2022
110 prop_binSR3Mu_Run3_bin6_ttZ_2023
111 CMS_eff_e_trigger_2017
112 CMS_eff_m_trigger_2016preVFP
113 prop_binSR1E2Mu_Run3_bin5
114 CMS_B2G25013_Norm_ZZ_13TeV
115 prop_binSR3Mu_Run3_bin6_ttZ_2022EE
116 CMS_B2G25013_Norm_tZq_13TeV
117 CMS_B2G25013_Norm_conversion_SR3Mu_13TeV
118 prop_binSR3Mu_Run3_bin6_ttZ_2023BPix
119 CMS_scale_j_2023
120 CMS_btag_fixedWP_light_uncorrelated_2018
−2 −1 0 1 2
−0.5 0 0.5
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



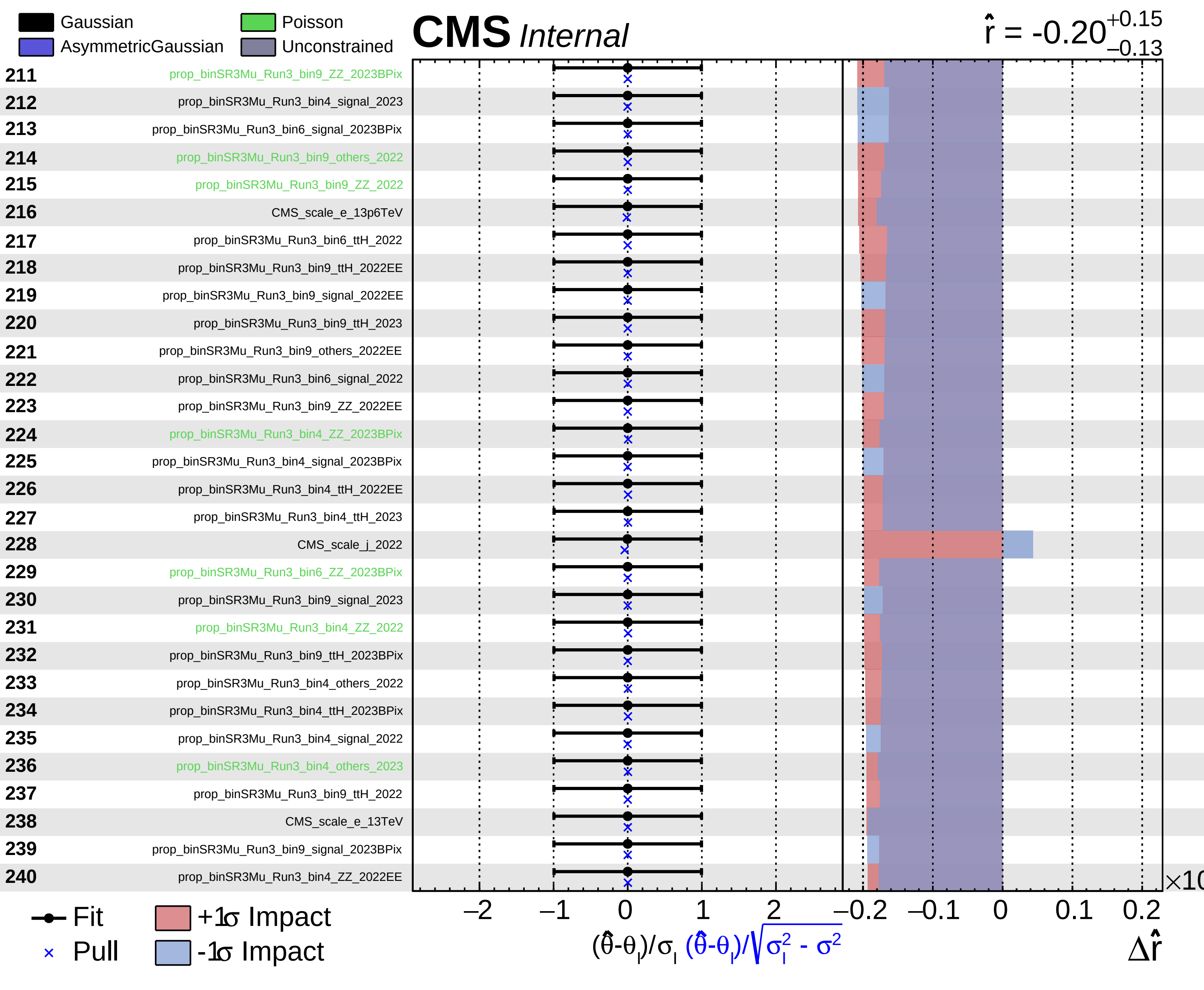
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = -0.20 +0.15 −0.13
121 ps_isr_13TeV
122 prop_binSR3Mu_Run3_bin11
123 prop_binSR1E2Mu_Run3_bin0
124 pdf_13p6TeV
125 CMS_eff_m_trigger_2016postVFP
126 prop_binSR3Mu_Run3_bin4_WZ_2023
127 CMS_btag_fixedWP_light_uncorrelated_2016postVFP
128 prop_binSR3Mu_Run3_bin9_ttW_2023BPix
129 prop_binSR1E2Mu_Run2_bin7
130 CMS_btag_fixedWP_light_uncorrelated_2017
131 prop_binSR3Mu_Run3_bin9_WZ_2023BPix
132 prop_binSR3Mu_Run2_bin3
133 CMS_scale_m_13p6TeV
134 ps_isr_13p6TeV
135 CMS_btag_fixedWP_bc_uncorrelated_2018
136 CMS_B2G25013_Norm_conversion_SR1E2Mu_2016preVFP
137 prop_binSR3Mu_Run3_bin6_ttZ_2022
138 lumi_13TeV_1516_I
139 CMS_eff_e_trigger_2016preVFP
140 prop_binSR3Mu_Run3_bin6_tZq_2022EE
141 prop_binSR3Mu_Run3_bin9_ttZ_2023
142 CMS_eff_m_trigger_2023BPix
143 prop_binSR3Mu_Run3_bin1
144 prop_binSR3Mu_Run3_bin6_tZq_2023
145 CMS_scale_j_2023BPix
146 prop_binSR3Mu_Run3_bin6_tZq_2023BPix
147 CMS_B2G25013_Norm_conversion_SR3Mu_13p6TeV
148 CMS_btag_fixedWP_light_uncorrelated_2016preVFP
149 prop_binSR3Mu_Run3_bin12
150 CMS_res_j_2016preVFP
−2
−1
0
1
2
−0.5
0
0.5
×10
Fit
+1σ Impact
Pull
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



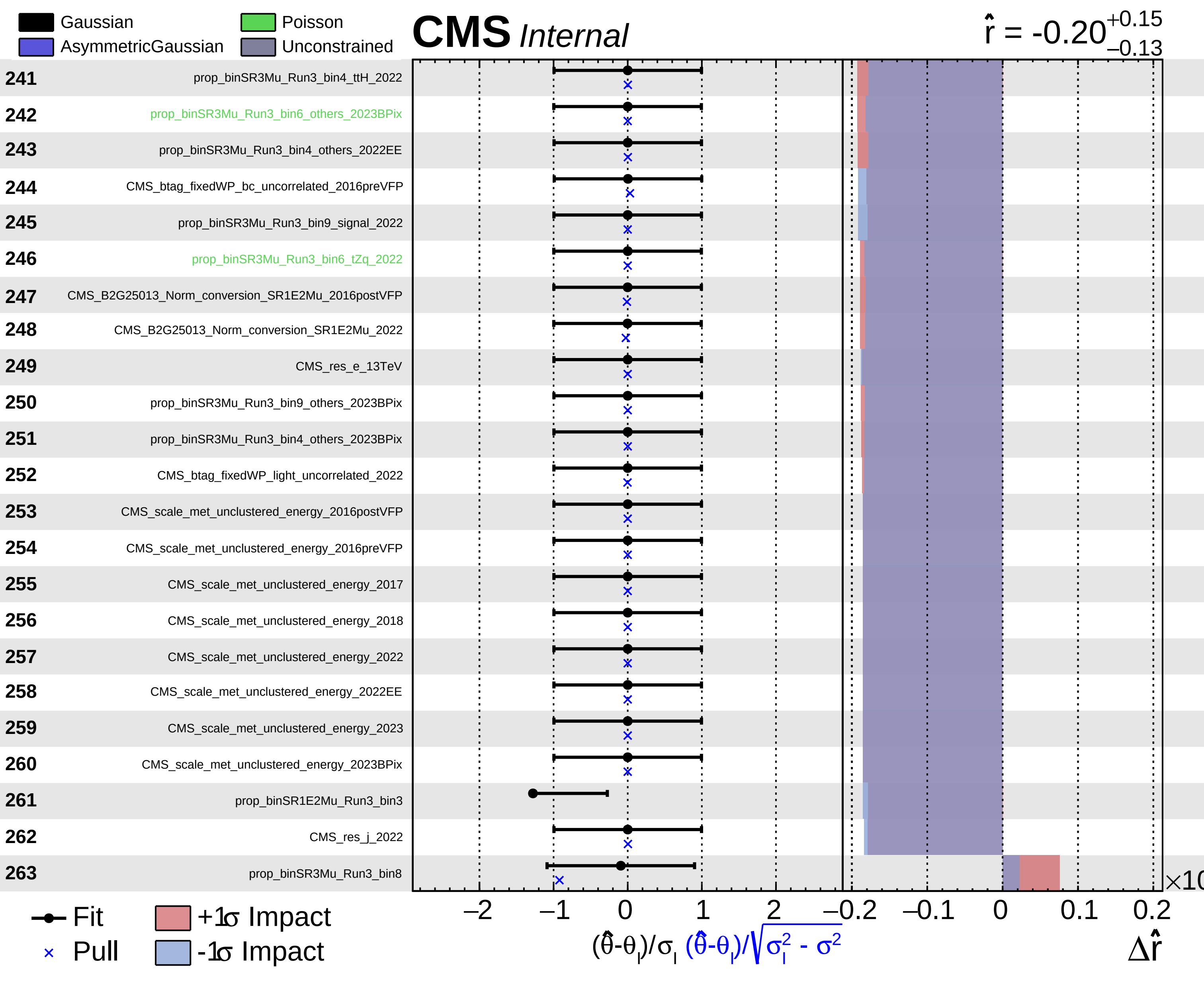
Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = -0.20^{+0.15}_{-0.13}$
151 prop_binSR3Mu_Run3_bin9_ttZ_2023BPix
152 CMS_eff_e_trigger_2016postVFP
153 lumi_13TeV_151617_l
154 prop_binSR3Mu_Run3_bin9_tZq_2023
155 prop_binSR3Mu_Run3_bin6_others_2023
156 prop_binSR3Mu_Run3_bin9_ttZ_2022EE
157 CMS_eff_m_trigger_2022
158 prop_binSR3Mu_Run3_bin4_ttW_2023BPix
159 prop_binSR3Mu_Run3_bin9_ttZ_2022
160 CMS_B2G25013_Norm_tZq_13p6TeV
161 CMS_res_j_2018
162 prop_binSR3Mu_Run3_bin4_WZ_2022EE
163 prop_binSR3Mu_Run3_bin9_WZ_2022EE
164 prop_binSR3Mu_Run3_bin4_ttZ_2022EE
165 CMS_B2G25013_Norm_conversion_SR1E2Mu_2022EE
166 prop_binSR3Mu_Run3_bin9_tZq_2023BPix
167 CMS_l1_prefiring_2016preVFP
168 CMS_B2G25013_Norm_conversion_SR1E2Mu_2023BPix
169 CMS_btag_fixedWP_light_uncorrelated_2023
170 prop_binSR3Mu_Run3_bin9_tZq_2022EE
171 CMS_btag_fixedWP_bc_uncorrelated_2016postVFP
172 prop_binSR3Mu_Run3_bin4_ttZ_2023
173 CMS_l1_prefiring_2018
174 CMS_B2G25013_Norm_ZZ_13p6TeV
175 CMS_res_e_13p6TeV
176 prop_binSR3Mu_Run3_bin4_tZq_2022EE
177 prop_binSR3Mu_Run3_bin4_ttZ_2022
178 CMS_btag_fixedWP_bc_uncorrelated_2017
179 CMS_B2G25013_Norm_conversion_SR1E2Mu_2023
180 prop_binSR3Mu_Run3_bin0
−2 −1 0 1 2
−0.2 0 0.2
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Internal
$\hat{r} = -0.20^{+0.15}_{-0.13}$
181 CMS_btag_fixedWP_light_uncorrelated_2022EE
182 CMS_B2G25013_Norm_conversion_SR1E2Mu_2017
183 prop_binSR3Mu_Run3_bin9_tZq_2022
184 prop_binSR3Mu_Run3_bin6_others_2022EE
185 CMS_l1_prefiring_2016postVFP
186 CMS_res_j_2023BPix
187 prop_binSR3Mu_Run3_bin4_ttW_2022
188 CMS_res_j_2016postVFP
189 prop_binSR1E2Mu_Run3_bin6
190 prop_binSR3Mu_Run3_bin4_tZq_2023BPix
191 prop_binSR3Mu_Run3_bin6_ZZ_2023
192 prop_binSR3Mu_Run3_bin6_ttH_2022EE
193 prop_binSR3Mu_Run3_bin4_nonprompt_2023
194 CMS_l1_prefiring_2017
195 prop_binSR3Mu_Run3_bin4_tZq_2022
196 prop_binSR3Mu_Run3_bin6_ttH_2023
197 prop_binSR3Mu_Run3_bin4_tZq_2023
198 CMS_res_j_2022EE
199 prop_binSR3Mu_Run3_bin6_ZZ_2022EE
200 prop_binSR3Mu_Run3_bin9_ZZ_2023
201 prop_binSR3Mu_Run3_bin6_signal_2022EE
202 prop_binSR3Mu_Run3_bin9_others_2023
203 prop_binSR3Mu_Run3_bin6_ttH_2023BPix
204 prop_binSR3Mu_Run3_bin6_signal_2023
205 CMS_res_j_2023
206 prop_binSR3Mu_Run3_bin6_ZZ_2022
207 CMS_btag_fixedWP_light_uncorrelated_2023BPix
208 prop_binSR3Mu_Run3_bin4_signal_2022EE
209 prop_binSR3Mu_Run3_bin6_others_2022
210 prop_binSR3Mu_Run3_bin4_ZZ_2023
−2
−1
0
1
2
−0.2
0
0.2
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
$\hat{r} = -0.20^{+0.15}_{-0.13}$
211 prop_binSR3Mu_Run3_bin9_ZZ_2023BPix
212 prop_binSR3Mu_Run3_bin4_signal_2023
213 prop_binSR3Mu_Run3_bin6_signal_2023BPix
214 prop_binSR3Mu_Run3_bin9_others_2022
215 prop_binSR3Mu_Run3_bin9_ZZ_2022
216 CMS_scale_e_13p6TeV
217 prop_binSR3Mu_Run3_bin6_ttH_2022
218 prop_binSR3Mu_Run3_bin9_ttH_2022EE
219 prop_binSR3Mu_Run3_bin9_signal_2022EE
220 prop_binSR3Mu_Run3_bin9_ttH_2023
221 prop_binSR3Mu_Run3_bin9_others_2022EE
222 prop_binSR3Mu_Run3_bin6_signal_2022
223 prop_binSR3Mu_Run3_bin9_ZZ_2022EE
224 prop_binSR3Mu_Run3_bin4_ZZ_2023BPix
225 prop_binSR3Mu_Run3_bin4_signal_2023BPix
226 prop_binSR3Mu_Run3_bin4_ttH_2022EE
227 prop_binSR3Mu_Run3_bin4_ttH_2023
228 CMS_scale_j_2022
229 prop_binSR3Mu_Run3_bin6_ZZ_2023BPix
230 prop_binSR3Mu_Run3_bin9_signal_2023
231 prop_binSR3Mu_Run3_bin4_ZZ_2022
232 prop_binSR3Mu_Run3_bin9_ttH_2023BPix
233 prop_binSR3Mu_Run3_bin4_others_2022
234 prop_binSR3Mu_Run3_bin4_ttH_2023BPix
235 prop_binSR3Mu_Run3_bin4_signal_2022
236 prop_binSR3Mu_Run3_bin4_others_2023
237 prop_binSR3Mu_Run3_bin9_ttH_2022
238 CMS_scale_e_13TeV
239 prop_binSR3Mu_Run3_bin9_signal_2023BPix
240 prop_binSR3Mu_Run3_bin4_ZZ_2022EE
−2
−1
0
1
2
−0.2
−0.1
0
0.1
0.2
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$



Gaussian
Poisson
AsymmetricGaussian
Unconstrained
CMS Internal
r̂ = -0.20 +0.15 −0.13
241 prop_binSR3Mu_Run3_bin4_ttH_2022
242 prop_binSR3Mu_Run3_bin6_others_2023BPix
243 prop_binSR3Mu_Run3_bin4_others_2022EE
244 CMS_btag_fixedWP_bc_uncorrelated_2016preVFP
245 prop_binSR3Mu_Run3_bin9_signal_2022
246 prop_binSR3Mu_Run3_bin6_tZq_2022
247 CMS_B2G25013_Norm_conversion_SR1E2Mu_2016postVFP
248 CMS_B2G25013_Norm_conversion_SR1E2Mu_2022
249 CMS_res_e_13TeV
250 prop_binSR3Mu_Run3_bin9_others_2023BPix
251 prop_binSR3Mu_Run3_bin4_others_2023BPix
252 CMS_btag_fixedWP_light_uncorrelated_2022
253 CMS_scale_met_unclustered_energy_2016postVFP
254 CMS_scale_met_unclustered_energy_2016preVFP
255 CMS_scale_met_unclustered_energy_2017
256 CMS_scale_met_unclustered_energy_2018
257 CMS_scale_met_unclustered_energy_2022
258 CMS_scale_met_unclustered_energy_2022EE
259 CMS_scale_met_unclustered_energy_2023
260 CMS_scale_met_unclustered_energy_2023BPix
261 prop_binSR1E2Mu_Run3_bin3
262 CMS_res_j_2022
263 prop_binSR3Mu_Run3_bin8
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$